Geospatial RAP Tool: Developer & Administrator Guide

This guide is intended for developers, IT administrators, and maintainers of the Geospatial RAP Tool. End-users should refer to the User Manual available within the application.

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1. Project Structure

The repository follows a modular Shiny architecture.

- Root directory: main scripts (`app.R`, `global.R`, `config.json`)
- `R/`: core logic and modules
- `data/`: optimized spatial layers (`.rds`)
- `build_data/`: preprocessing scripts
- `www/`: UI assets and documentation

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2. Running the App Locally

1. Open the project in RStudio
2. Open `app.R`
3. Run:

```
shiny::runApp()
```

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3. Deployment

Option A: shinyapps.io

- Install `rsconnect`
- Deploy using `rsconnect::deployApp()`

Option B: Self-hosted

- Shiny Server (Linux)
- Docker containerization

Option C: Desktop (Electron)

- Bundles R + app into standalone executable
- No user setup required

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4. Configuration

4.1 Geographic Configuration (`config.json`)

Defines:

- Administrative levels
- Mapping between CSV and spatial layers
- Spatial IDs and labels

Supports multi-country setups without modifying core code.

4.2 UI Customization (`config.css`)

Controls:

- Layout and colors
- Sidebar appearance
- Map background

4.3 Offline Spatial Data

Optimized `.rds` layers are stored in:

data/

To generate them:

build_data/optimize_spatial_data.R

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5. Data Pipeline

The application:

- Reads API-POP CSV files
- Standardizes column names
- Pads geographic identifiers
- Aggregates data
- Performs spatial joins

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6. Spatial Analysis Engine

- Global Moran's I
- Local Moran's I (LISA)
- Handles sparse data using safe defaults

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7. Troubleshooting

- Missing polygons → check spatial IDs
- Empty maps → verify configuration
- Performance issues → ensure `.rds` usage

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8. Documentation

- User Manual: `www/docs/Documentation.pdf`
- Spanish version: `HELP_ES.md`